SDS8102: Advanced Mathematical Statistics

Lecture 4: Quasi-Likelihood, KL and Hellinger distances, EM algorithm

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Lecture 4: Beyond Maximum Likelihood

Maximum likelihood starts from a fully specified statistical model

$$\{P_\theta : \theta \in \Theta\}.$$

The likelihood uses the complete distribution of the observed data:

$$L(\theta; X) = f_\theta(X).$$

But in many problems, specifying or using the full likelihood may be unnecessary or difficult.

For example:

- we may trust only certain features of the model;
- nuisance parameters may make the likelihood complicated;
- conditioning may remove irrelevant nuisance parameters;
- latent variables may make direct likelihood optimization difficult.

This motivates several extensions of the likelihood principle.

Learning Objectives

By the end of this lecture, you should be able to

1. Understand the basic idea of quasi-likelihood.
2. Use conditioning to eliminate nuisance parameters.
3. Interpret maximum likelihood through KL divergence.
4. Compare distributions using KL and Hellinger distances.
5. Understand why latent variables make likelihood optimization difficult.
6. Derive the basic EM algorithm.

When Is a Full Likelihood Necessary?

Suppose the response Y has conditional distribution

$$Y \mid X = x \sim P_{\theta,x}.$$

A full likelihood requires specifying the entire distribution

$$p_\theta(y \mid x).$$

But sometimes our scientific model specifies only certain features of the distribution.

For example, suppose Y is the number of hospital visits for a patient with covariates $X = x$. We may be willing to model

$$\mathbb{E}[Y \mid X = x] = \mu_\theta(x)$$

and assume a mean–variance relationship such as

$$\mathrm{Var}(Y \mid X = x) = \phi\, \mu_\theta(x),$$

without specifying the entire distribution of $Y \mid X = x$.

Can we still estimate θ without specifying $p_\theta(y \mid x)$?

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From Likelihood Scores to Quasi-Scores

For a regular likelihood, the score is

$$S(\theta) = \frac{\partial}{\partial\theta} \log L(\theta).$$

The MLE is typically obtained by solving

$$S(\widehat{\theta}) = 0.$$

This suggests a broader strategy:

Instead of specifying a full likelihood, construct an **estimating equation** that behaves like a likelihood score.

Such an equation is called a **quasi-score**.

Thus quasi-likelihood connects naturally to the Z-estimation framework from the previous lecture.

Quasi-Likelihood

Suppose Y has mean and variance

$$\mathbb{E}[Y] = \mu, \qquad \mathrm{Var}(Y) = \phi V(\mu),$$

where $V(\mu)$ is a known variance function and $\phi > 0$ is a dispersion parameter.

A quasi-likelihood $Q(\mu; y)$ is defined through

$$\boxed{\frac{\partial Q(\mu; y)}{\partial \mu} = \frac{y - \mu}{\phi V(\mu)}.}$$

Equivalently,

$$Q(\mu; y) = \int^{\mu} \frac{y - t}{\phi V(t)}\, dt$$

up to an additive constant.

The key point is that this construction requires only the **mean–variance relationship**, not a full probability model.

Why Is This a Reasonable Estimating Equation?

Consider the quasi-score

$$U(\mu; Y) = \frac{Y - \mu}{\phi V(\mu)}.$$

At the true mean,

$$\mathbb{E}[U(\mu; Y)] = \frac{\mathbb{E}[Y] - \mu}{\phi V(\mu)} = 0.$$

Thus the true parameter solves the population estimating equation

$$\boxed{\mathbb{E}[U(\mu; Y)] = 0.}$$

Its empirical analogue can therefore be used to estimate μ.

This is exactly the logic of a Z-estimator:

$$\text{population equation} \quad \longrightarrow \quad \text{empirical equation.}$$

Example: Constant Variance

Suppose

$$\mathbb{E}[Y] = \mu, \qquad \mathrm{Var}(Y) = \sigma^2,$$

but we do not assume that Y is Gaussian.

Then

$$V(\mu) = 1, \qquad \phi = \sigma^2.$$

The quasi-score is

$$U(\mu; Y) = \frac{Y - \mu}{\sigma^2}.$$

For independent observations, the estimating equation is

$$\sum_{i=1}^{n} \frac{Y_i - \mu}{\sigma^2} = 0 \Longrightarrow \boxed{\widehat{\mu} = \overline{Y}}$$

The sample mean therefore arises without assuming a Normal distribution.

Likelihood versus Quasi-Likelihood

The two ideas should not be confused.

	Likelihood	Quasi-likelihood
Requires full distribution?	Yes	No
Uses mean structure?	Yes	Yes
Uses variance structure?	Implicitly	Explicitly
Must be a probability model?	Yes	No

A quasi-likelihood is **not** generally the logarithm of a probability density.

Its purpose is to retain useful likelihood-like estimating equations when a full distributional specification is unavailable or undesirable.

Nuisance Parameters

Suppose the parameter can be written as

$$\theta = (\psi, \lambda),$$

where ψ is the parameter of interest and λ is a **nuisance parameter**.

The full likelihood is

$$L(\psi, \lambda; X).$$

Although we are interested only in ψ, estimation may require us to deal with λ as well.

This raises a natural question:

$$\boxed{\text{Can we remove the nuisance parameter before estimation?}}$$

One possibility is to condition on a statistic whose distribution captures the nuisance parameter.

Conditional Likelihood

Write the data as (Y, T) and suppose their joint density factors as

$$p_{\psi,\lambda}(y, t) = p_\psi(y \mid t)\, p_{\psi,\lambda}(t),$$

where the conditional distribution of Y given T does not depend on the nuisance parameter λ.

Then the **conditional likelihood** for ψ is

$$\boxed{L_c(\psi; y \mid t) = p_\psi(y \mid T = t).}$$

We can estimate ψ by maximizing

$$\widehat{\psi} \in \arg\max_\psi L_c(\psi; y \mid t).$$

Conditioning has removed λ from the likelihood.

A Simple Example

Suppose X and Y are independent with

$$X \sim \text{Poisson}(\lambda), \qquad Y \sim \text{Poisson}(\lambda\theta),$$

where $\theta > 0$ is the parameter of interest and $\lambda > 0$ is a nuisance parameter.

The joint likelihood depends on both:

$$L(\theta, \lambda; x, y) \propto e^{-\lambda(1+\theta)} \lambda^{x+y} \theta^{y}.$$

Direct maximum likelihood therefore requires dealing with λ.

Can we condition on something that removes it?

Conditioning Removes the Nuisance Parameter

Let

$$T = X + Y.$$

A standard property of independent Poisson variables gives

$$Y \mid T = t \sim \text{Binomial}\left(t, \frac{\theta}{1+\theta}\right).$$

Crucially, λ has disappeared.

Thus the conditional likelihood is

$$L_c(\theta; y \mid t) = \binom{t}{y} \left(\frac{\theta}{1+\theta}\right)^{y} \left(\frac{1}{1+\theta}\right)^{t-y}.$$

Ignoring terms independent of θ,

$$\boxed{L_c(\theta) \propto \frac{\theta^y}{(1+\theta)^t}.}$$

Conditional MLE

The conditional log-likelihood is

$$\ell_c(\theta) = y \log \theta - t \log(1+\theta) + C.$$

Differentiating,

$$\ell_c'(\theta) = \frac{y}{\theta} - \frac{t}{1+\theta}.$$

Setting the score equal to zero gives

$$\frac{y}{\widehat{\theta}} = \frac{t}{1+\widehat{\theta}}.$$

Since $t = x + y$,

$$\boxed{\widehat{\theta}_c = \frac{y}{x}.}$$

We have estimated the relative rate θ without estimating the nuisance rate λ.

Why Conditional Likelihood Is Useful

Conditional likelihood can simplify inference by eliminating nuisance parameters.

The basic strategy is

$$\boxed{\text{Full model} \rightarrow \text{condition on } T \rightarrow \text{nuisance disappears} \rightarrow \text{estimate parameter of interest.}}$$

But conditioning comes with a potential cost:

- we deliberately discard information contained in the marginal distribution of T;
- a useful conditioning statistic may not exist;
- the resulting conditional likelihood need not contain all information about the parameter of interest.

Thus conditioning is particularly attractive when it removes difficult nuisance parameters cleanly.

Why Should Maximum Likelihood Work?

Suppose the data are generated from a true distribution

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} P_{\theta_0}.$$

Maximum likelihood chooses

$$\widehat{\theta} \in \arg\max_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^{n} \log p_\theta(X_i).$$

By the law of large numbers, for each fixed θ,

$$\frac{1}{n} \sum_{i=1}^{n} \log p_\theta(X_i) \longrightarrow \mathbb{E}_{\theta_0}[\log p_\theta(X)].$$

So, at the population level, maximum likelihood tries to maximize

$$\theta \longmapsto \mathbb{E}_{\theta_0}[\log p_\theta(X)].$$

Why should this objective be maximized at θ_0?

Kullback–Leibler Divergence

Definition: KL Divergence

For distributions P and Q with densities p and q, the Kullback–Leibler divergence from Q to P is

$$\boxed{D_{\mathrm{KL}}(P\|Q) = \int p(x)\log\frac{p(x)}{q(x)}\,dx.}$$

Equivalently,

$$D_{\mathrm{KL}}(P\|Q) = \mathbb{E}_P\left[\log\frac{p(X)}{q(X)}\right].$$

For discrete distributions, the integral is replaced by a sum.

Basic Properties of KL Divergence

KL divergence satisfies

$$D_{\mathrm{KL}}(P\|Q) \geq 0.$$

Moreover,

$$D_{\mathrm{KL}}(P\|Q) = 0 \quad \Longleftrightarrow \quad P = Q \quad \text{a.e.}$$

However, KL divergence is generally not symmetric:

$$D_{\mathrm{KL}}(P\|Q) \neq D_{\mathrm{KL}}(Q\|P).$$

Therefore KL divergence is **not a metric**.

In particular, it does not represent ordinary Euclidean distance between probability distributions.

Why Is KL Divergence Nonnegative?

Using Jensen's inequality,

$$\begin{aligned} -D_{\mathrm{KL}}(P\|Q) &= \mathbb{E}_P\left[\log \frac{q(X)}{p(X)}\right] \\ &\leq \log \mathbb{E}_P\left[\frac{q(X)}{p(X)}\right] \\ &= \log \int q(x)\,dx \\ &= 0. \end{aligned}$$

Therefore

$$\boxed{D_{\mathrm{KL}}(P\|Q) \geq 0.}$$

Equality in Jensen's inequality occurs precisely when

$$\frac{q(X)}{p(X)}$$

is constant P-almost surely, which implies $P = Q$.

Maximum Likelihood and KL Divergence

Consider

$$D_{\mathrm{KL}}(P_{\theta_0} \| P_\theta).$$

By definition,

$$\begin{aligned} D_{\mathrm{KL}}(P_{\theta_0} \| P_\theta) &= \mathbb{E}_{\theta_0}\left[\log \frac{p_{\theta_0}(X)}{p_\theta(X)}\right] \\ &= \mathbb{E}_{\theta_0}[\log p_{\theta_0}(X)] - \mathbb{E}_{\theta_0}[\log p_\theta(X)]. \end{aligned}$$

The first term does not depend on θ.

Therefore,

$$\boxed{\arg\max_\theta \mathbb{E}_{\theta_0}[\log p_\theta(X)] = \arg\min_\theta D_{\mathrm{KL}}(P_{\theta_0} \| P_\theta).}$$

MLE as Empirical KL Minimization

Maximum likelihood maximizes

$$\frac{1}{n}\sum_{i=1}^{n}\log p_\theta(X_i),$$

which approximates

$$\mathbb{E}_{\theta_0}[\log p_\theta(X)].$$

At the population level, maximizing expected log-likelihood is equivalent to minimizing

$$D_{\mathrm{KL}}(P_{\theta_0}\|P_\theta).$$

Hence the MLE can be viewed as trying to find the model distribution closest to the data-generating distribution in KL divergence:

$$\boxed{\text{MLE} \quad \approx \quad \text{KL projection onto the statistical model.}}$$

What If the Model Is Wrong?

Suppose the true distribution P_0 does not belong to the model:

$$P_0 \notin \{P_\theta : \theta \in \Theta\}.$$

At the population level, maximum likelihood targets

$$\theta^* \in \arg\max_{\theta} \mathbb{E}_{P_0}[\log p_\theta(X)].$$

Equivalently,

$$\boxed{\theta^* \in \arg\min_{\theta \in \Theta} D_{\mathrm{KL}}(P_0 \| P_\theta).}$$

Thus θ^* is the **KL projection** of the true distribution onto the statistical model.

The MLE can therefore have a meaningful population target even when the model is misspecified.

Example: KL Between Normal Distributions

Consider

$$P = N(\mu_0, \sigma^2), \qquad Q = N(\mu, \sigma^2).$$

A direct calculation gives

$$\boxed{D_{\mathrm{KL}}(P \| Q) = \frac{(\mu - \mu_0)^2}{2\sigma^2}.}$$

Thus, when the variance is fixed, KL divergence is quadratic in the difference between the means.

In particular,

$$D_{\mathrm{KL}}(P \| Q)$$

becomes larger when the mean difference is large relative to the noise level σ.

Is KL Really a Distance?

KL divergence gives a useful notion of discrepancy between distributions, but

$$D_{\mathrm{KL}}(P\|Q) \neq D_{\mathrm{KL}}(Q\|P)$$

in general.

It can also be infinite when P places probability mass where Q assigns zero probability.

So it is natural to ask:

$$\boxed{\text{Can we define a symmetric and bounded discrepancy?}}$$

One important answer is the **Hellinger distance**.

Hellinger Distance

Let P and Q have densities p and q with respect to a common dominating measure.

Definition: Hellinger Distance

The squared Hellinger distance between P and Q is

$$\boxed{H^2(P,Q) = \frac{1}{2}\int \left(\sqrt{p(x)} - \sqrt{q(x)}\right)^2 dx.}$$

The Hellinger distance is

$$H(P,Q) = \sqrt{H^2(P,Q)}.$$

Unlike KL divergence, Hellinger distance is symmetric:

$$H(P,Q) = H(Q,P).$$

An Equivalent Representation

Expanding the square,

$$H^2(P, Q) = \frac{1}{2} \int (p + q - 2\sqrt{pq})\, dx$$
$$= 1 - \int \sqrt{p(x)q(x)}\, dx.$$

Therefore,

$$\boxed{H^2(P, Q) = 1 - \int \sqrt{p(x)q(x)}\, dx.}$$

The quantity

$$\rho(P, Q) = \int \sqrt{p(x)q(x)}\, dx$$

is called the **Hellinger affinity**.

Thus

$$H^2(P, Q) = 1 - \rho(P, Q).$$

Basic Properties

The Hellinger distance satisfies

$$0 \leq H(P,Q) \leq 1.$$

Moreover,

$$H(P,Q) = 0 \quad \Longleftrightarrow \quad P = Q.$$

Unlike KL divergence,

$$H(P,Q) = H(Q,P),$$

and Hellinger distance satisfies the triangle inequality.

Therefore,

$$\boxed{H(P,Q) \text{ is a genuine metric on probability distributions.}}$$

The Geometry Behind Hellinger Distance

Associate each density p with its square root

$$p \quad \longmapsto \quad \sqrt{p}.$$

Since

$$\int (\sqrt{p})^2 dx = \int p(x)\, dx = 1,$$

every $\sqrt{p}$ lies on the unit sphere of the space L^2.

Moreover,

$$H(P, Q) = \frac{1}{\sqrt{2}} \left\| \sqrt{p} - \sqrt{q} \right\|_{L^2}.$$

Thus Hellinger distance is simply Euclidean distance between square-root densities in L^2, up to a scaling factor.

$\boxed{\text{Probability distributions acquire a genuine geometric structure.}}$

Example: Hellinger Distance Between Normals

Consider

$$P = N(\mu_1, \sigma^2), \qquad Q = N(\mu_2, \sigma^2).$$

Their Hellinger affinity is

$$\rho(P, Q) = \exp\left\{-\frac{(\mu_1 - \mu_2)^2}{8\sigma^2}\right\}.$$

Therefore,

$$\boxed{H^2(P, Q) = 1 - \exp\left\{-\frac{(\mu_1 - \mu_2)^2}{8\sigma^2}\right\}.}$$

As $|\mu_1 - \mu_2|$ increases relative to σ, the two distributions become increasingly separated.

KL versus Hellinger

Both quantify discrepancy between probability distributions, but they have different properties.

Property	KL	Hellinger
Nonnegative	Yes	Yes
Zero iff $P = Q$	Yes	Yes
Symmetric	No	Yes
Triangle inequality	No	Yes
Bounded	No	Yes
Metric	No	Yes

KL divergence arises naturally from **likelihood**.

Hellinger distance provides a particularly convenient **metric geometry** on probability distributions.

A Parametric Family as a Geometric Object

Consider a smooth parametric family

$$\mathcal{P} = \{P_\theta : \theta \in \Theta\}.$$

Changing θ moves us through a family of probability distributions:

$$\theta \longmapsto P_\theta.$$

KL divergence and Hellinger distance quantify how far apart nearby members of this family are.

For a small perturbation h,

$$P_\theta \qquad \text{and} \qquad P_{\theta+h}$$

become increasingly similar as $h \to 0$.

A remarkable fact is that, locally, these discrepancies have an approximately **quadratic geometry**.

$\boxed{\text{The local geometry will lead us to Fisher information.}}$

When Maximum Likelihood Becomes Difficult

Maximum likelihood asks us to solve

$$\widehat{\theta} \in \arg\max_{\theta} \ell(\theta; X).$$

Sometimes the likelihood is easy to write down but difficult to optimize.

A common reason is the presence of **latent variables**.

Suppose the complete data would be

$$(X, Z),$$

but we observe only X.

The observed-data likelihood is then

$$p_\theta(X) = \sum_z p_\theta(X, z)$$

for discrete Z, or an analogous integral for continuous Z.

The logarithm of this sum can make direct optimization difficult.

Observed Data versus Complete Data

Suppose Z is an unobserved latent variable.

The observed-data log-likelihood is

$$\ell(\theta; X) = \log p_\theta(X) = \log \sum_z p_\theta(X, z).$$

The **complete-data log-likelihood** is

$$\ell_c(\theta; X, Z) = \log p_\theta(X, Z).$$

In many models,

$$\boxed{\ell_c(\theta; X, Z) \text{ is much easier to optimize than } \ell(\theta; X).}$$

If only we knew Z, maximum likelihood would be straightforward.

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The EM algorithm replaces the missing Z by its conditional distribution given the observed data.

The Basic Idea of EM

Suppose our current parameter estimate is

$$\theta^{(t)}.$$

Since Z is unknown, compute its conditional distribution

$$p_{\theta^{(t)}}(Z \mid X).$$

Then average the complete-data log-likelihood with respect to this distribution:

$$\boxed{Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{\theta^{(t)}}\left[\log p_\theta(X, Z) \mid X\right].}$$

The next parameter value is obtained by maximizing this expected complete-data log-likelihood.

The EM Algorithm

Starting from an initial value $\theta^{(0)}$, repeat:

Example: E-Step

Compute

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{\theta^{(t)}} \left[\log p_\theta(X, Z) \mid X \right].$$

Example: M-Step

Update

$$\boxed{\theta^{(t+1)} \in \arg\max_\theta Q(\theta \mid \theta^{(t)}).}$$

Continue until the parameter values or likelihood values stabilize.

Why Does EM Increase the Likelihood?

For any distribution $q(z)$,

$$\begin{aligned}\log p_\theta(X) &= \log \sum_z p_\theta(X, z) \\ &= \log \sum_z q(z) \frac{p_\theta(X, z)}{q(z)}.\end{aligned}$$

By Jensen's inequality,

$$\log p_\theta(X) \geq \sum_z q(z) \log \frac{p_\theta(X, z)}{q(z)}.$$

Thus every choice of q gives a lower bound on the observed-data log-likelihood.

EM repeatedly constructs and maximizes such a lower bound.

The E-Step Makes the Bound Tight

Choose

$$q^{(t)}(z) = p_{\theta^{(t)}}(z \mid X).$$

With this choice, the lower bound is tight at the current parameter:

$$\mathcal{L}(q^{(t)}, \theta^{(t)}) = \log p_{\theta^{(t)}}(X).$$

The M-step chooses $\theta^{(t+1)}$ to increase this lower bound.

Therefore,

$$\boxed{\ell(\theta^{(t+1)}; X) \geq \ell(\theta^{(t)}; X).}$$

Thus EM produces a sequence of nondecreasing likelihood values.

This does **not** imply convergence to the global maximum.

Example: A Gaussian Mixture Model

Suppose

$$X_1, \ldots, X_n \stackrel{\text{iid}}{\sim} \pi N(\mu_1, \sigma^2) + (1-\pi) N(\mu_2, \sigma^2).$$

The observed density is

$$p_\theta(x) = \pi \phi(x; \mu_1, \sigma^2) + (1-\pi)\phi(x; \mu_2, \sigma^2).$$

Hence

$$\ell(\theta) = \sum_{i=1}^{n} \log \left[\pi \phi(X_i; \mu_1, \sigma^2) + (1-\pi)\phi(X_i; \mu_2, \sigma^2) \right].$$

The logarithm of a sum makes direct maximization inconvenient.

Introduce Latent Cluster Labels

Introduce

$$Z_i = \begin{cases} 1, & X_i \text{ comes from component 1,} \\ 0, & X_i \text{ comes from component 2.} \end{cases}$$

If Z_i were observed, the complete-data likelihood would be

$$\prod_{i=1}^{n} \left[\pi\phi(X_i; \mu_1, \sigma^2)\right]^{Z_i} \left[(1-\pi)\phi(X_i; \mu_2, \sigma^2)\right]^{1-Z_i}.$$

The complete-data log-likelihood is therefore

$$\begin{aligned} \ell_c(\theta) = \sum_{i=1}^{n} \Big[& Z_i \log \pi + (1 - Z_i) \log(1 - \pi) \\ & + Z_i \log \phi(X_i; \mu_1, \sigma^2) \\ & + (1 - Z_i) \log \phi(X_i; \mu_2, \sigma^2) \Big]. \end{aligned}$$

E-Step: Compute cluster membership weights

Since Z_i is unobserved, compute

$$\gamma_i^{(t)} = \mathbb{E}[Z_i \mid X_i, \theta^{(t)}].$$

Because Z_i is binary,

$$\gamma_i^{(t)} = P(Z_i = 1 \mid X_i, \theta^{(t)}).$$

Bayes' rule gives

$$\boxed{\gamma_i^{(t)} = \frac{\pi^{(t)}\phi(X_i; \mu_1^{(t)}, \sigma^2)}{\pi^{(t)}\phi(X_i; \mu_1^{(t)}, \sigma^2) + (1 - \pi^{(t)})\phi(X_i; \mu_2^{(t)}, \sigma^2)}}.$$

The quantity $\gamma_i^{(t)}$ is called the **responsibility** of component 1 for observation i.

M-Step: Update the Parameters

Replace the unknown labels Z_i by their responsibilities $\gamma_i^{(t)}$.

Maximizing the expected complete-data likelihood gives

$$\boxed{\pi^{(t+1)} = \frac{1}{n}\sum_{i=1}^{n} \gamma_i^{(t)}.}$$

The component means become weighted averages:

$$\boxed{\mu_1^{(t+1)} = \frac{\sum_i \gamma_i^{(t)} X_i}{\sum_i \gamma_i^{(t)}},}$$

and

$$\boxed{\mu_2^{(t+1)} = \frac{\sum_i (1-\gamma_i^{(t)}) X_i}{\sum_i (1-\gamma_i^{(t)})}.}$$

EM for the Gaussian Mixture

The algorithm alternates between two operations:

E-step	:	Estimate soft cluster memberships,
M-step	:	Re-estimate parameters using those memberships.

Graphically,

$$\theta^{(t)} \longrightarrow \{\gamma_i^{(t)}\}_{i=1}^{n} \longrightarrow \theta^{(t+1)} \longrightarrow \cdots$$

Unlike hard clustering, the latent labels are not immediately assigned as 0 or 1.

Instead, each observation receives a probability of belonging to each component.

What EM Guarantees—and What It Does Not

EM guarantees that the observed-data likelihood does not decrease:

$$\ell(\theta^{(t+1)}) \geq \ell(\theta^{(t)}).$$

However, EM does not generally guarantee convergence to the global maximum.

Possible issues include:

- sensitivity to initialization;
- convergence to local maxima or stationary points;
- slow convergence near a solution;
- multiple equivalent solutions in mixture models.

Thus EM converts a difficult optimization problem into a sequence of simpler optimization problems, but it does not eliminate the nonconvexity of the original problem.

The Bigger Picture

This week we have seen several approaches to estimation:

MoM	:	match empirical and population moments
M-estimation	:	optimize an empirical objective
Z-estimation	:	solve an estimating equation
MLE	:	maximize likelihood
Quasi-likelihood	:	use mean–variance structure
Conditional likelihood	:	condition away nuisance parameters
EM	:	handle latent variables iteratively

KL and Hellinger divergence give us tools for understanding the geometry of the underlying statistical models.